

A Tour of AI Skepticism Through the Lens of the Theory of Computation

Workshop Slide Deck

Anonymous Author(s)

Workshop Info

Join via Zoom:

<https://utoronto.zoom.us/j/7550080142>

Format: Informal workshop / discussion

No expectations — but questions, discussion, and feedback are welcome!

Pre-requisites: Familiarity with P vs NP helps; otherwise, follow-up reading recommended

Outline

Introduction

The Church-Turing Thesis: What IS a Computer?

The Frame Problem

The Chomsky Hierarchy

NP-Completeness and Learnability

Discussion

Introduction

AI Skepticism: A Long Tradition

Skepticism about machines recreating human intelligence has deep roots:

- **Ada Lovelace** (1840s)
- **Noam Chomsky** — linguistics and cognition
- **Jerry Fodor** — cognitive science
- **John Searle** — Chinese Room argument
- **Roger Penrose** — physics and consciousness

Much of this skepticism can be framed in terms of the **Theory of Computation** – but often isn't.

This talk: fill that gap.

Advances in LLMs have prompted new skeptical arguments:

- **Bender & Koller (2020)** — “Climbing towards NLU”
- **van Rooij et al. (2024)** — “Reclaiming AI”

Today: reframe these arguments in *precise mathematical terms*

Ada Lovelace's Objection

Ada Lovelace (1843) on Charles Babbage's Analytical Engine:

"The Analytical Engine has no pretensions to originate anything. It can do whatever we know how to order it to perform."



John Searle (1980):

The Argument

- A person in a room follows a set of rules to manipulate Chinese symbols
- The person does not understand Chinese
- Therefore, syntactic manipulation does not imply semantic understanding

Implication: Symbol manipulation (computation) is not sufficient for understanding

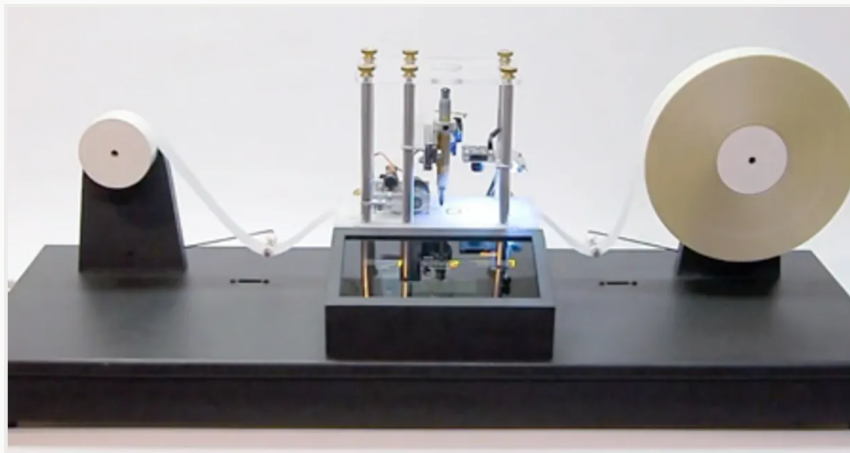
Computational perspective:

- A LARGE room
- An ENORMOUS amount of energy
- An UNBELIEVABLY COMPLEX set of rules
- Does your intuition about normal rooms really matter here?

The Church-Turing Thesis: What IS a Computer?

- A implementation of what most people think of as a “computer”
- A tape of infinite length divided into cells
- A “head” that can read and write symbols on the tape
- The “head” can move left or right one cell at a time
- A finite table of rules that determine the head’s next action based on the current internal state and the symbol being read

Turing Machines



<https://spectrum.ieee.org/032610-diy-turing-machine>

On the board: a simple Turing machine

The Church-Turing Thesis

Thesis: Any function that is “effectively computable” can be computed by a Turing machine (equivalently, Python on a CPU)

Implications for AI

- If cognition is computable $\Rightarrow$ Turing machines can simulate it
- But: is cognition computable?

Extended Church-Turing Thesis: Any “reasonable” model of computation can be simulated by a Turing machine with at most polynomial slowdown. (e.g., quantum computers don’t violate this)

What Turing Machines cannot do

Turing's Halting Problem and Goedel's Incompleteness Theorem, in VS Code

Halting problem

On the board `https://www.youtube.com/watch?v=r24f3RmMjHs&t=1s`

Goedel's Incompleteness Theorem

On the board `https://www.youtube.com/watch?v=91j\_JbIghM`

Penrose's Objection

Roger Penrose (Physics Nobel Laureate):

Human mathematical insight transcends algorithmic computation

The Argument (roughly)

- Gödel's incompleteness: no consistent formal system can prove all true statements
- Humans can “see” truths that no algorithm can prove
- Therefore: the mind is not algorithmic

Relies on quantum effects in microtubules (controversial, not widely accepted)

Quick review:

- **P:** Problems solvable in polynomial time
- **NP:** Problems verifiable in polynomial time
- **NP-complete:** Hardest problems in NP

The question: Is $P = NP$?

If $P \neq NP$, many problems (e.g., theorem proving) are intractable in the worst case.

Reductions: Transforming one problem into another

- If problem A reduces to problem B, solving B efficiently implies solving A efficiently
- NP-complete problems: all NP problems reduce to them
- Implication: if any NP-complete problem is in P, then $P = NP$

On the board: Halting Problem and Godel Incompleteness Theorem reductions

- Proofs on the board, following Aaronson notes style, using Python as model of computation.

Goedel's Incompleteness Theorem

- Discussion
- J. Ellenberg's counterargument

Reduction: Example

Example: 3-SAT reduces to CLIQUE

- 3-SAT: Given a boolean formula in CNF with 3 literals per clause, is there a satisfying assignment?
- CLIQUE: Given a graph and integer k , is there a clique of size k ?
- Reduction: Construct a graph from the 3-SAT instance such that a k -clique corresponds to a satisfying assignment

Implication: If we can solve CLIQUE efficiently, we can solve 3-SAT efficiently

P, NP, and NP-Completeness: Examples

On the board: examples of problems in P, NP, and NP-complete

Learning as search: Many learning problems can be framed as search problems

- Finding a hypothesis consistent with training data
- Optimizing model parameters
- If these problems are NP-hard, efficient learning may be impossible
- Generally: arbitrary functions are **not** efficiently learnable (Valiant 1984)

Open question: Are there efficient algorithms for learning human-like intelligence?

The Frame Problem

Jerry Fodor: How does a cognitive system determine what is *relevant*?

The Problem

- Updating beliefs requires knowing what changes and what stays the same
- Potentially infinite number of things to check
- How do humans do this efficiently?

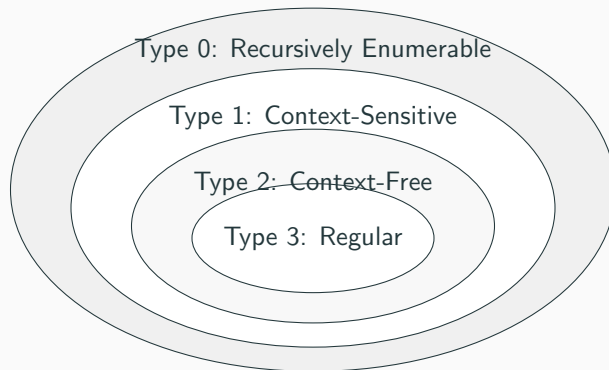
Computational framing: Relevance determination seems to require global computation
Computational framing: Exponential time to compute: need to try all possible frames?

Computational perspective:

- Naive approach: check all possible changes
- Exponential blowup in number of factors
- Humans seem to avoid this combinatorial explosion
- Is there a polynomial-time algorithm for relevance?
- Transformers determine the frame problem one token at a time (but maybe miss something (but so do humans?))

The Chomsky Hierarchy

The Chomsky Hierarchy



Chomsky's claim: Natural language requires at least context-sensitive grammars; statistical models miss the hierarchical structure.

On the board: examples of languages in each Chomsky hierarchy level. Arguments for why English is high on the level.

- Language is generative, rule-governed, hierarchically structured
- Statistical pattern matching is fundamentally different
- “Colorless green ideas sleep furiously” — grammatical but meaningless
- LLMs capture surface patterns, not underlying competence

Computational question: Can finite data reveal infinite generative capacity?

Gold's Theorem (1967):

Result

No algorithm can learn all recursively enumerable languages from positive examples alone

Implication: Learning human language may require more than just exposure to positive examples (e.g., innate constraints, negative evidence)

NP-Completeness and Learnability

P vs NP: Quick Review

- **P:** Problems solvable in polynomial time
- **NP:** Problems verifiable in polynomial time
- **NP-complete:** Hardest problems in NP

The question: Is $P = NP$?

If $P \neq NP$, many optimization and learning problems are intractable in the worst case.

“Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data”

Key Argument

- Meaning is grounded in communicative intent
- Form alone (text) cannot convey meaning
- LLMs learn form, not meaning
- “Octopus test”: can a system truly understand?

Computational angle: What can be learned from form alone?

- Bender and Koller (2020) slides: "https://docs.google.com/presentation/d/17wP_0KEVSUqAtIg3aP4YWJAxJdZURxrOypgVPWwffbE/"
- Note: even if Physics argument is true, learning is not required to be tractable.

“Reclaiming AI as a Theoretical Tool for Cognitive Science”

Claim

Achieving human-like intelligence via learning from data is *intractable*

Critique (Guerzhoy 2024): The proof relies on unjustified assumptions about (input, output) distributions

What assumptions are needed for intractability results?

van Rooij et al. (2024), on the board.

Barriers to Intractability Proofs

From Guerzhoy (2024), “Barriers to Complexity-Theoretic Proofs that Achieving AGI Using Machine Learning is Intractable”:

- Worst-case complexity $\neq$ average-case complexity
- Distribution of real-world problems matters
- Current proof techniques may be insufficient

Open question: Can we prove (or disprove) fundamental limits on learning?

Discussion

Theory of Computation provides a lens for AI skepticism:

- **Church-Turing Thesis:** What is computable?
- **Penrose:** Is the mind algorithmic?
- **Frame Problem:** Relevance and tractability
- **Chomsky Hierarchy:** Formal language theory and linguistic competence
- **NP-completeness:** Tractability of learning

These are *mathematical* questions with *philosophical* implications.

- What can be learned from finite data?
- Does scalability resolve intractability concerns?
- Is “understanding” computationally definable?
- What would constitute a proof of AI (im)possibility?

Discussion welcome!

References

Books:

- Aaronson, S. *Quantum Computing Since Democritus*

Articles:

- Bender, E. & Koller, A. (2020). Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data. *ACL 2020*.
<https://aclanthology.org/2020.acl-main.463/>
- van Rooij, I. et al. (2024). Reclaiming AI as a Theoretical Tool for Cognitive Science. *Computational Brain and Behavior*.
<https://link.springer.com/article/10.1007/s42113-024-00217-5>

- Guerzhoy, M. (2024). Barriers to Complexity-Theoretic Proofs that Achieving AGI Using Machine Learning is Intractable.
<https://arxiv.org/abs/2411.06498>
- Guerzhoy, M. (2026). Towards Formalizing Skepticism of Language Models: A Taxonomy in the Language of the Theory of Computation. PhilML Workshop at ICML 2026 (non-archival).

Stanford Encyclopedia of Philosophy:

- The Frame Problem

<https://plato.stanford.edu/entries/frame-problem/>

Additional background:

- Fodor, J. — Lectures 4–6 on the Frame Problem
- <https://slate.com/human-interest/2005/03/the-romantic-s-favorite-mathematician.html>

Thank you!

Questions & Discussion